



Final Exam of Computer Programming II
University of Basrah
College of Computer Science and Information Technology
Department of Computer Information Systems
2024-2025/ Second Semester



Exam Duration: 2 Hours

Stage 1/ First Attempt

Date: 15-6-2025

- N.B. (1) The maximum mark for this paper is 35 [8+9+9+9]
(2) This exam has **THREE** pages. Make sure you have all the pages before you begin.
(3) Please read each question carefully before you start to answer it.
(4) You have to leave one question that have (9) Marks only.

Q1: A) Answer The following questions with True or False and Correct the error if False
(4 Marks)

1. This code finds the sum of the first n even numbers. <pre>sum = 0; a = 2; cout << "Enter n = "; cin >> n; for (int i = 1; i <= n; i++) { sum += a; a += 2; }</pre>	2. This program prints A in the first row, B in the second, and it continues. <pre>char ch = 'A'; int n = 5; for (int i = 1; i <= n; i++) { for (int j = 1; j <= i; j++) { cout << ch << " "; cout << endl; ch++; } }</pre> 2/9
3. This code adds numbers from 1 to 5. <pre>sum = 0; for (int i = 1; i < 5; i++) { sum += i; }</pre> False	4. This program prints a triangle. <pre>int n = 4; for (int i = 1; i <= n; i++) { for (int j = 1; j <= n; j++) { cout << "*" << " "; cout << endl; } }</pre> True

B) What is the output of the following C++ programs? (ONLY TWO)
(4 Marks)

<pre>(1) { int number, divisor; cout << "Enter the number: "; cin >> number; cout << "Enter the divisor: "; cin >> divisor; if (divisor == 0) cout << "Division by zero." ; else if (number % divisor == 0) cout << " is a multiple number "; else cout << " is not a multiple number"; return 0 ;}</pre>	<pre>(2) { int val = 2; switch (val) { case 1: cout << " case 1 "; break; case 2: cout << " case 2 "; case 3: cout << " case 3 "; default: cout << " default output "; } return 0 ;}</pre>
<pre>(3) { int num = 5; while (num <= 95) { cout << num << "\t"; num += 10; } cout << endl; return 0; }</pre>	<pre>(4) { int Y, K, result = 1; cout << "Enter the base (Y): "; cin >> Y; cout << "Enter the exponent (K): "; cin >> K; for (int i = 1; i <= K; i++) result *= Y; cout << "result=" << result; return 0;}</pre>

Q2: Answer the following:

(9 Marks)

A) Write a C++ program to read an unknown quantity of positive integer numbers and find their average. **Not:** Your program ends with the input 0. (5 Marks)

B) Write a suitable statement for the following (TWO ONLY): (4 Marks)

1. Write an Algorithm to find the Area of the Circle
2. Design a flowchart for the largest among three numbers the user enters.
3. Design a flowchart that generates even numbers between 1000 and 2000 and prints them in the standard output. It should also print the total sum.

B) What is the output of the following C++ programs? (ONLY TWO) (4 Marks)

(A) <pre> int n = 5; for (int i = 1; i <= n; i++) { for (int j = 1; j <= i; j++) { if (i % 2 == 1) cout << "1 "; else cout << "2 "; } cout << endl; } </pre>	(B) <pre> int sum=0; int a=99; int f=1; int n=4; for (int i=1;i<=n;i++){ sum=sum + f *a; f*=-1; a -= 11;} cout << " sum = " << sum; </pre>
(C) <pre> int n = 8; int a [] = { 18, 25, 36, 44, 12, 60, 75, 89 }; int x = a [0]; for (int i = 1; i < n; i++) if (a [i] < x) x = a [i]; cout << "The x number is: " << x; </pre>	(D) <pre> int table[3][3] = { {1, 2, 3}, {4, 5, 6}, {7, 8, 9} }; int result = 0; int rowIndex = 2; for (int c = 0; c < 3; c++) result += table[rowIndex][c]; cout << " result = " << result; </pre>

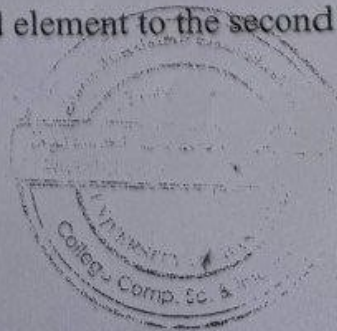
Q2: Answer the following:

(9 Marks)

A) write a C++ program with a function called max(). This function receives num1 and num2 as parameters and returns the maximum value (5 Marks)

B) Write a suitable statement for the following (TWO ONLY): (4 Marks)

1. Write only function mult() that accepts two floating-point, multiplies them, and returns the result.
2. Write only a function named square() that takes an integer, computes its square, and returns the result.
3. Write only function that copies the value of the third element to the second in an array.



6. What values will be printed of variables x and y in the following code execution:

```
int x = 2, y = 5;
y = x++ + ++x;
cout << ++y << "..." << x++;
```

- a. 6...3 b. 6...4 c. 6...5 d. 7...4

7. How many times is a do while loop guaranteed to loop?
- a. 0 b. Infinitely c. 1 d. Variable

8. Find out the error in following block of code:

```
If (x = 100)
cout << "x is 100";
```

- a. 100 should be enclosed in quotations " " .
b. There is no semicolon (;) at the end of first line.
c. Equals to operator mistake (==).
d. There are no mistakes.

9. I would like to insert comments in the source code. How can I apply comments?
- a. use // b. use " " c. use { } d. ;

10. Identify the correct example for a pre-decrement operator.
- a. --n b. n-- c. ++n d. n++

11. Which one of the following statements should be used to insert a new line in a C++ program?
- a. \t b. \n c. \b d. \r

12. Which of the following condition will be true if x=5, y=2?
- a. x < 4 || y > 4 b. !(x > 10 && y > 1) c. x > 4 && y > 4 d. None of them

(10 Marks)

Q2: Answer the following (Only Five)

1. Your company needs printers and provides money 1000\$ to buy as many of the same printer as possible, the price of the printers is stored in the variable price. Fill in the blank to calculate and output how much money will be left over after buying the printers.

```
int money=1000;
int price =145;
[ ] result=money [ ] price;
cout<< [ ] ;
```

2. What is the value of num after this code?

```
int num = 2;
int x = 3;
if (num >= 2)
{ if (x > 4)
    {num=4;}
  else
    {num=10;}
}
else
    {x=1;}
```


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Lecturers: Wid Akeel Awadh + Noor Mohammed

Date: 8-1-2023

19

30

Student Name:

محمود محمد

Q1: Write a C++ program that takes a 4-digit number as input and outputs the reversed number. For example, if the input is 1234, the output should be 4321. (7 marks)

~~include <iostream>~~
~~using namespace std;~~

int main()

{
int a=1, b=2, c=3, d=4;

cout <<

2

return 0;

}

Q2: Answer the following questions:

(15 marks)

A. Convert the expression to valid C++ code. $z = \frac{x+y}{x(3+y^2)}$	cout << z << endl; cout << (x+y)/(x*(3+y*y)) << endl;
B. What is the output of the following code segment: <pre>int x, y=8, z=9; x = z++/++y; cout << x << " " << y << " " << z << endl;</pre>	3 9 10

2/11

3/11

- int i = 5, j = 6, k = 7, n = 3;
cout << i + j * k - k % n;
- a. 43 b. 45 c. 47 d. 46

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Q3: Draw a flowchart to illustrate the following algorithm then write the program.

C. What is the output of the program below?

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << (n = 4) << endl;
    cout << (n == 4) << endl;
    cout << (n > 3) << endl;
    cout << (n = 0) << endl;
    cout << (n && 4) << endl;
    cout << (n || 4) << endl;
    cout << (!n) << endl;
    return 0; }
```

- ① 4 ✓
- ② 4 ✓
- ③ 4 ✓
- ④ 0 ✓
- ⑤ 0 ✓
- ⑥ 0 ✓
- ⑦ 0 ✓

D. How can it be corrected to do what is the programmer surely intended?

```
int n = 5;
if (n = 0) NOTE THE OPERATOR!!!
    cout << "n is zero" << "\n";
else
    cout << "n is not zero" << "\n";
cout << "The square of n is " << n * n << "\n";
```

if (n = 0) // Not the OPERATOR!!!
cout << "n is zero.\n";
else if
cout << "n is not zero.\n";
else
cout << "The square of n is " << n * n << "\n";

E. If a, b, c, d and e are of integer types and a=11, b=6, c=0, d=7, e=5. What will be the value of this expression:

$a+2 > b \ \&\& \ !c \ || \ a \ != \ d \ \&\& \ a-2 \leq e$

$a+2 > b \ \&\& \ !c \Rightarrow a=13, c=0;$
True out put 1
 $a \ != \ d \ \&\& \ a-2 \leq e \Rightarrow 11, 9 \leq 5$
True out put 1

21/11/24
3/11/24

$z = \frac{x}{3 + y^2}$

B. What is the output of the following code segment:

```
int x, y=8, z=9;
x = z++/++y;
cout << x << " " << y << " " << z << endl;
```

1 9 10

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13

30

Student Name: A2

(15 marks)

Q1: Choose the correct answer:

1. What will `!(1 && 0) || 1` evaluate to?

a. 1

b. 0

c. error

d. no one

2. Is this code correct? `int n=10; int x=5; int n=x;`

a. Yes

b. No

3. What will be the results of the following code:

```
int x=5, y=10;
if (x < y) cout << "1";
x = x * 2;
if (x == y) cout << "2";
```

a. 1

b. 2

c. 10

d. 50

4. What are the values of m?

```
int k=20, int m=4;
```

```
k -= ++m + 5;
```

```
m = k + m;
```

a. 16

b. 15

c. 18

d. 20

5. What will be the results of the following code:

5. Identify the correct example for a pre-increment operator.

a. ++n

b. n++

c. +n

d. -n

6. Which of the following loops is best when we know the number of iterations?

a. While loop

b. Do while loop

c. For loop

d. All of above

7. Identify the logical AND operator.

a. ||

b. &&

c. &

d. AND

8. Choose the type of loop which is guaranteed to execute at-least once?

a. While loop

b. Do while loop

c. For loop

d. All of above

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C. What is the output of the program below? <pre>#include <iostream> using namespace std; int main() { int n; cout << (n = 4) << endl; cout << (n == 4) << endl; cout << (n > 3) << endl; cout << (n = 0) << endl; cout << (n && 4) << endl; cout << (n 4) << endl; cout << (!n) << endl; return 0; }</pre>	① 4 ✓ ② 4 ✓ ③ 4 ✓ ④ 0 ✓ ⑤ 0 ✓ ⑥ 0 ✓ ⑦ 0 ✓
D. How can it be corrected to do what is the programmer surely intended? <pre>int n = 5; if (n = 0) { cout << "n is zero" << ".\n"; } else { cout << "n is not zero" << ".\n"; cout << "The square of n is " << n * n << ".\n"; }</pre>	if (n == 0) // Not the OPERATOR!!! cout << "n is zero.\n"; else if cout << "n is not zero.\n"; else cout << "the square of n is " << n * n << ".\n";
E. If a, b, c, d and e are of integer types and a=11, b=6, c=0, d=7, e=5. What will be the value of this expression: $a+2 > b \ \&\& \ !c \ \ a \ != \ d \ \&\& \ a-2 < = e$	$a+2 > b \ 8 > 6 \ 8 \ != \ 0 \Rightarrow a=13, c=0$ + true output 1 $a \ != \ d \ 11 \ != \ 7 \ 11-2 < = e \Rightarrow 9 < = 5$ True output 1

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Q2: Write a program to print the following shape:

(8 marks)

```
1 1 1 1 4
2 2 2 2 8
3 3 3 3 16
4 4 4 4 32
```


Q3: Chosen one only.

(10 Mark)

A) Write a C++ program to input marks of N students in two subjects, CS101 and CS103. The program must be print the success rate in each subject.

✓ B) Write Program to Solve the Equation ($Y = X^2 + X + 1 / X$) Where X Value Between 2 And 7

Q4: Answer the following: (ONE ONLY)

(10 Mark)

✓ A) Write a program using the nested While to print the product and sum the numbers from 0 to 5 (five times).

B) write a C++ program to read the temperature at Celsius and display a suitable message according to the temperature below:

- Temperature is smaller than 0 (Freezing weather)
- Temperature 0-10 (Very Cold weather)
- Temperature 10-20 (Cold weather)
- Temperature 20-30 (Normal in Temp)
- Temperature 30-40 (It's Hot)
- Temperature is greater than 40 (It's Very Hot)

Examiner

Noor Al-Moosawi/ Ali Abdulrazzaq

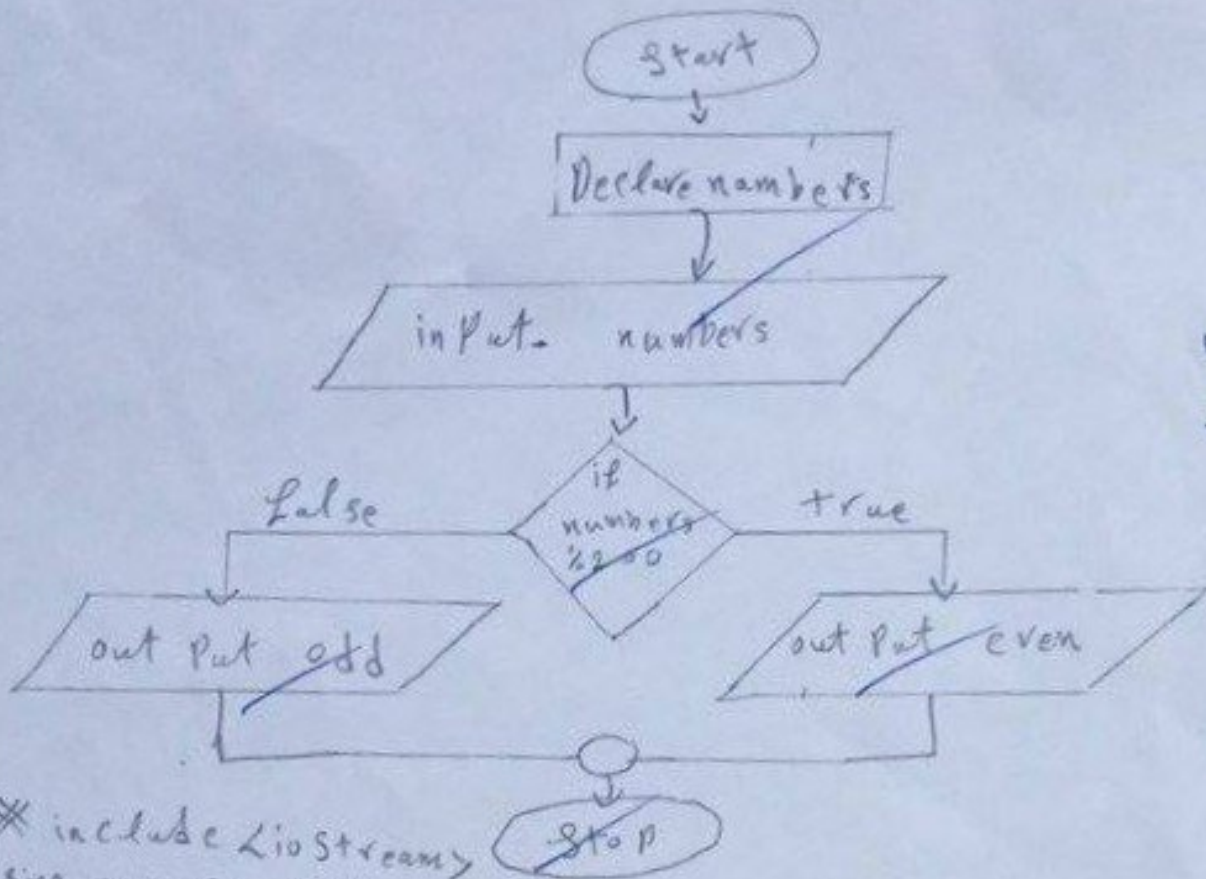
Head of the Department
Prof. Dr. Haider

M. Al- Mashheedi

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Date: 8-1-2024

Q3: Draw a flowchart to illustrate the process of determining whether a given number is even or odd then write program in C++ language to solve this task using switch .. case statement. (8 marks)



```
* include <iostream>
using namespace std;
int main()
{
    int number;
    cout << "Enter The number " << endl;
    cin >> number;
    switch (number % 2)
    {
        case 0:
            cout << "the number is even";
            break;
        case 1:
            cout << "the number is odd";
            break;
        default:
            cout << "No result";
    }
    return 0;
}
```

3/11

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9. Find the output of the following program.

```
Int main() {  
    Float a = 5;  
    switch(a)  
    {  
        Case 5: cout << "Interviewbit";  
        Default: cout << "Scaler";  
    }
```

- a. Interviewbit b. Scaler c. Error d. No one

10. Find the output of the following program.

```
Int main()  
{  
    int a = 10, b, c;  
    b = a++;  
    c = a;  
    cout << a << " " << b << " " << c << endl;  
}
```

- a. 10 11 11 b. 11 11 11 c. 11 10 11 d. 10 10 10

11. Identify the correct escape sequence used for the new line.

- a. \t b. \n c. endl d. b and c

12. Which of the following is used for comments in C++?

- a. /* comment */ b. // comment */ c. // comment d. both a and c

13. What is the output of the code segment below?

```
int n = 3;  
while (n >= 0)  
{  
    cout << n * n << endl;  
    --n;  
}
```

- a. 9 4 1 b. 8 6 2 c. Error d. No one

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١٤. What is the output when the following code fragment is executed?

```
int i = 5, j = 6, k = 7, n = 3;  
cout << i + j * k - k % n << endl;
```

- a. 46 b. 1 c. 76 d. No one

١٥. What is the output when the following code fragment is executed?

```
int found = 0, count = 5;  
if (!found || --count == 0)  
    cout << "danger" << endl;  
cout << "count = " << count << endl;
```

- a. Danger b. Count c. 4 d. 0

Q2: Write a C++ program to evaluate the factorial of a positive number.

(7 marks)